Precision Operational Amplifier

FEATURES

- *Guaranteed* Low Offset Voltage
  - LT1001AM 15μV max
  - LT1001C 60μV max
- *Guaranteed* Low Drift
  - LT1001AM 0.6μV/°C max
  - LT1001C 1.0μV/°C max
- *Guaranteed* Low Bias Current
  - LT1001AM 2nA max
  - LT1001C 4nA max
- *Guaranteed* CMRR
  - LT1001AM 114dB min
  - LT1001C 110dB min
- *Guaranteed* PSRR
  - LT1001AM 110dB min
  - LT1001C 106dB min
- Low Power Dissipation
  - LT1001AM 75mW max
  - LT1001C 80mW max
- Low Noise 0.3μV_{p-p}

APPLICATIONS

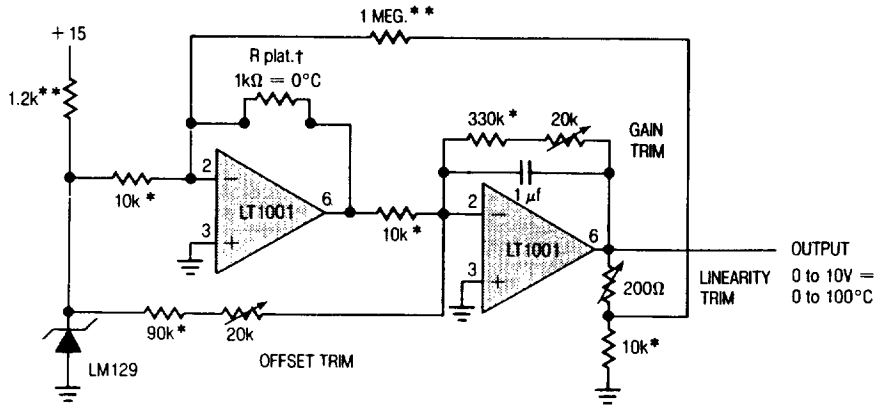
- Thermocouple amplifiers
- Strain gauge amplifiers
- Low level signal processing
- High accuracy data acquisition

DESCRIPTION

The LT1001 significantly advances the state-of-the-art of precision operational amplifiers. In the design, processing, and testing of the device, particular attention has been paid to the optimization of the entire distribution of several key parameters. Consequently, the specifications of the lowest cost, commercial temperature device, the LT1001C, have been dramatically improved when compared to equivalent grades of competing precision amplifiers.

Essentially, the input offset voltage of all units is less than 50μV (see distribution plot below). This allows the LT1001AM/883 to be specified at 15μV. Input bias and offset currents, common-mode and power supply rejection of the LT1001C offer guaranteed performance which were previously attainable only with expensive, selected grades of other devices. Power dissipation is nearly halved compared to the most popular precision op amps, without adversely affecting noise or speed performance. A beneficial by-product of lower dissipation is decreased warm-up drift. Output drive capability of the LT1001 is also enhanced with voltage gain guaranteed at 10 mA of load current. For similar performance in a dual precision op amp, with guaranteed matching specifications, see the LT1002. Shown below is a platinum resistance thermometer application.

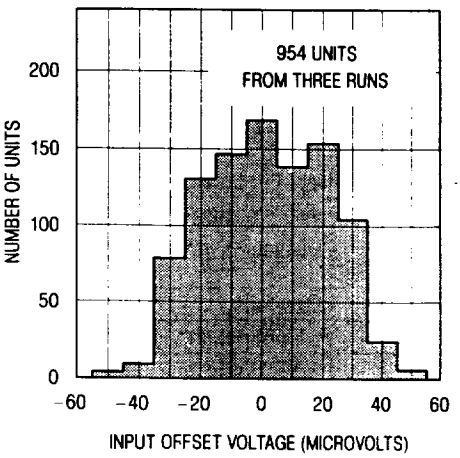
Linearized Platinum Resistance Thermometer with ±0.025°C Accuracy Over 0 to 100°C



* ULTRONIX 105A WIREWOUND
** 1% FILM
† PLATINUM RTD 118MF (ROSEMOUNT, INC.)

‡ Trim sequence: trim offset (0°C = 1000.0Ω), trim linearity (35°C = 1138.7Ω), trim gain (100°C = 1392.6Ω). Repeat until all three points are fixed with ±.025°C.

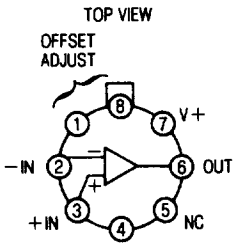
Typical Distribution of Offset Voltage
V_S = ±15V, T_A = 25°C



ABSOLUTE MAXIMUM RATINGS

Supply Voltage ..... ±22V
Differential Input Voltage ..... ±30V
Input Voltage ..... ±22V
Output Short Circuit Duration ..... Indefinite
Operating Temperature Range
LT1001AM/LT1001M ..... −55°C to 150°C
LT1001AC/LT1001C ..... 0°C to 125°C
Storage: All Devices ..... −65°C to 150°C
Lead Temperature (Soldering, 10 sec.) ..... 300°C

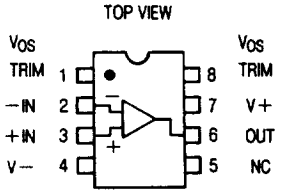
PACKAGE/ORDER INFORMATION



H PACKAGE
METAL CAN

ORDER PART NUMBER

LT1001AMH/883
LT1001MH
LT1001ACH
LT1001CH



J8 PACKAGE
8 PIN HERMETIC DIP

N8 PACKAGE
8 PIN PLASTIC DIP

LT1001AMJ8/883
LT1001MJ8
LT1001ACJ8
LT1001CJ8

LT1001ACN8
LT1001CN8

ELECTRICAL CHARACTERISTICS

$V_S = \pm 15V$, $T_A = 25°C$, unless otherwise noted

SYMBOL	PARAMETER	CONDITIONS	LT1001AM/883 LT1001AC MIN	LT1001AM/883 LT1001AC TYP	LT1001AM/883 LT1001AC MAX	LT1001M/LT1001C MIN	LT1001M/LT1001C TYP	LT1001M/LT1001C MAX	UNITS
V_{OS}	Input Offset Voltage	Note 1 LT1001AM/883		7	15		18	60	μV
		LT1001AC		10	25				
$\frac{\Delta V_{OS}}{\Delta \text{ Time}}$	Long Term Input Offset Voltage Stability	Notes 2 and 3		0.2	1.0		0.3	1.5	μV/month
I_{OS}	Input Offset Current			0.3	2.0		0.4	3.8	nA
I_b	Input Bias Current			±0.5	±2.0		±0.7	±4.0	nA
e_n	Input Noise Voltage	0.1Hz to 10Hz (Note 2)		0.3	0.6		0.3	0.6	μV_{p-p}
e_n	Input Noise Voltage Density	f_o = 10Hz (Note 5)		10.3	18.0		10.5	18.0	$nV\sqrt{Hz}$
		f_o = 1000Hz (Note 2)		9.6	11.0		9.8	11.0	
A_{VOL}	Large Signal Voltage Gain	$R_L \geq 2k\Omega$, $V_o = \pm 12V$	450	800		400	800		V/mV
		$R_L \geq 1k\Omega$, $V_o = \pm 10V$	300	500		250	500		
CMRR	Common Mode Rejection Ratio	$V_{CM} = \pm 13V$	114	126		110	126		dB
PSRR	Power Supply Rejection Ratio	$V_S = \pm 3V$ to $\pm 18V$	110	123		106	123		dB
R_{in}	Input Resistance Differential Mode	(Note 4)	30	100		15	80		MΩ
	Input Voltage Range		±13	±14		±13	±14		V
V_{OUT}	Maximum Output Voltage Swing	$R_L \geq 2k\Omega$	±13	±14		±13	±14		V
		$R_L \geq 1k\Omega$	±12	±13.5		±12	±13.5		
S_R	Slew Rate	$R_L \geq 2k\Omega$ (Note 4)	0.1	0.25		0.1	0.25		V/μs
GBW	Gain-Bandwidth Product	(Note 4)	0.4	0.8		0.4	0.8		MHz
P_d	Power Dissipation	No load		46	75		48	80	mW
		No load, $V_s = \pm 3V$		4	6		4	8	

See Notes on page 3.

ELECTRICAL CHARACTERISTICS $V_S = \pm 15V$, $-55°C \leq T_A \leq 125°C$, unless otherwise noted

SYMBOL	PARAMETER	CONDITIONS		LT1001AM/883 MIN	LT1001AM/883 TYP	LT1001AM/883 MAX	LT1001M MIN	LT1001M TYP	LT1001M MAX	UNITS
V_{OS}	Input Offset Voltage		●		30	60		45	160	μV
$\frac{\Delta V_{OS}}{\Delta \text{ Temp}}$	Average Offset Voltage Drift		●		0.2	0.6		0.3	1.0	μV/°C
I_{OS}	Input Offset Current		●		0.8	4.0		1.2	7.6	nA
I_B	Input Bias Current		●		±1.0	±4.0		±1.5	±8.0	nA
A_{VOL}	Large Signal Voltage Gain	$R_L \geq 2k\Omega$, $V_O = \pm 10V$	●	300	700		200	700		V/mV
CMRR	Common Mode Rejection Ratio	$V_{CM} = \pm 13V$	●	110	122		106	120		dB
PSRR	Power Supply Rejection Ratio	$V_S = \pm 3$ to $\pm 18V$	●	104	117		100	117		dB
	Input Voltage Range		●	±13	±14		±13	±14		V
V_{OUT}	Output Voltage Swing	$R_L \geq 2k\Omega$	●	±12.5	±13.5		±12.0	±13.5		V
P_d	Power Dissipation	No load	●		55	90		60	100	mW

2

$V_S = \pm 15V$, $0°C \leq T_A \leq 70°C$, unless otherwise noted

SYMBOL	PARAMETER	CONDITIONS		LT1001AC MIN	LT1001AC TYP	LT1001AC MAX	LT1001C MIN	LT1001C TYP	LT1001C MAX	UNITS
V_{OS}	Input Offset Voltage		●		20	60		30	110	μV
$\frac{\Delta V_{OS}}{\Delta \text{ Temp}}$	Average Offset Voltage Drift		●		0.2	0.6		0.3	1.0	μV/°C
I_{OS}	Input Offset Current		●		0.5	3.5		0.6	5.3	nA
I_B	Input Bias Current		●		±0.7	±3.5		±1.0	±5.5	nA
A_{VOL}	Large Signal Voltage Gain	$R_L \geq 2k\Omega$, $V_O = \pm 10V$	●	350	750		250	750		V/mV
CMRR	Common Mode Rejection Ratio	$V_{CM} = \pm 13V$	●	110	124		106	123		dB
PSRR	Power Supply Rejection Ratio	$V_S = \pm 3V$ to $\pm 18V$	●	106	120		103	120		dB
	Input Voltage Range		●	±13	±14		±13	±14		V
V_{OUT}	Output Voltage Swing	$R_L \geq 2k\Omega$	●	±12.5	±13.8		±12.5	±13.8		V
P_d	Power Dissipation	No load	●		50	85		55	90	mW

The ● denotes the specifications which apply over the full operating temperature range.

Note 1: Offset voltage for the LT1001AM/883 and LT1001AC are measured after power is applied and the device is fully warmed up. All other grades are measured with high speed test equipment, approximately 1 second after power is applied. The LT1001AM/883 receives 168 hr. burn-in at 125°C. or equivalent.

Note 2: This parameter is tested on a sample basis only.

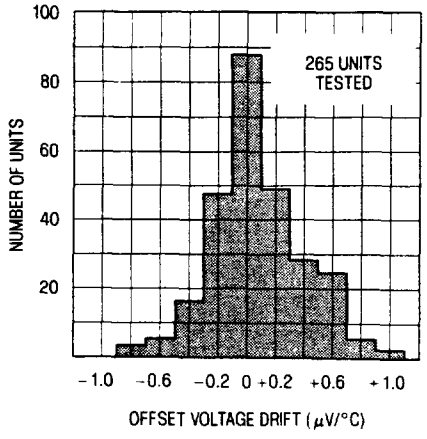
Note 3: Long Term Input Offset Voltage Stability refers to the averaged trend line of V_{os} versus Time over extended periods after the first 30 days of operation. Excluding the initial hour of operation, changes in V_{os} during the first 30 days are typically 2.5μV.

Note 4: Parameter is guaranteed by design.

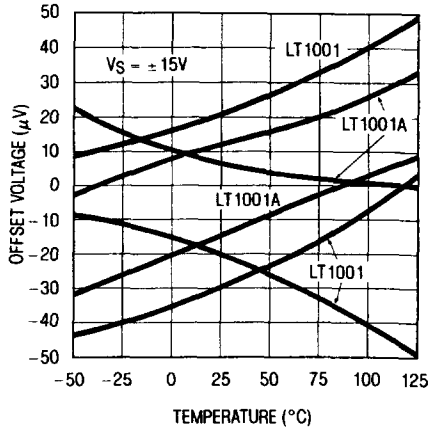
Note 5: 10Hz noise voltage density is sample tested on every lot. Devices 100% tested at 10Hz are available on request.

TYPICAL PERFORMANCE CHARACTERISTICS

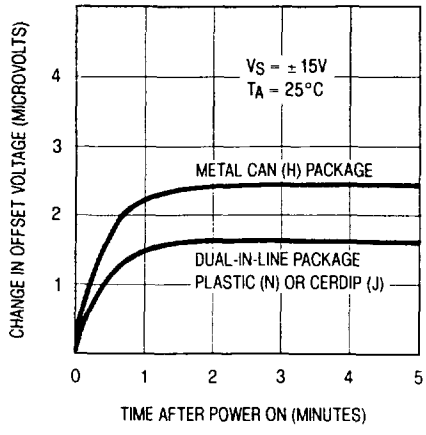
Typical Distribution of Offset Voltage Drift with Temperature



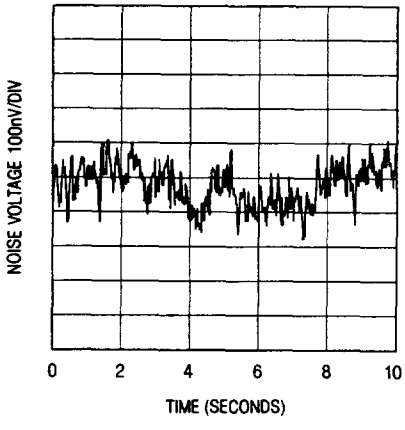
Offset Voltage Drift with Temperature of Representative Units



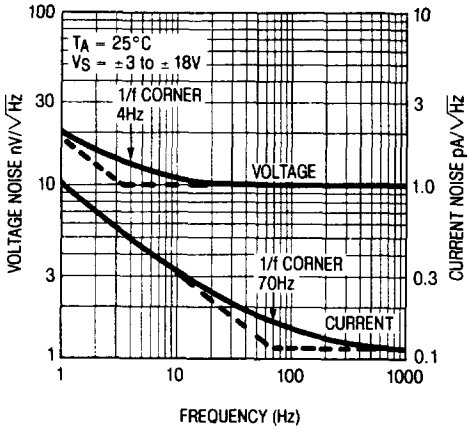
Warm-Up Drift



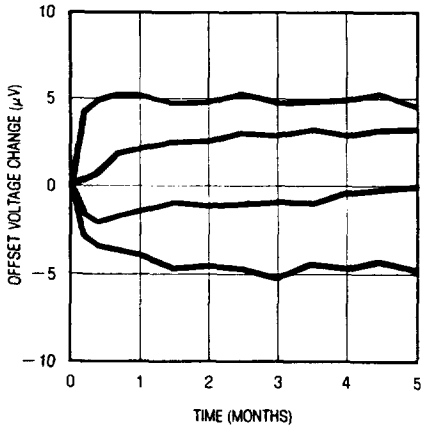
0.1Hz to 10Hz Noise



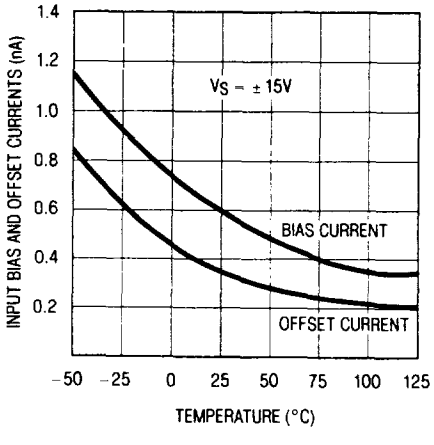
Noise Spectrum



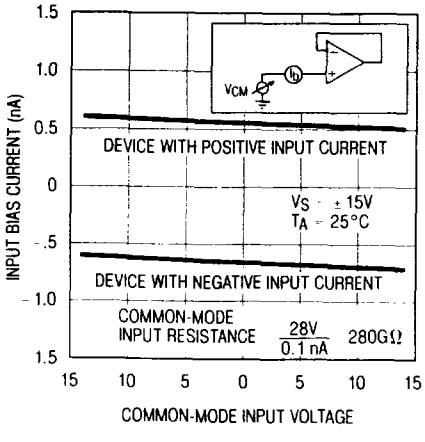
Long Term Stability of Four Representative Units



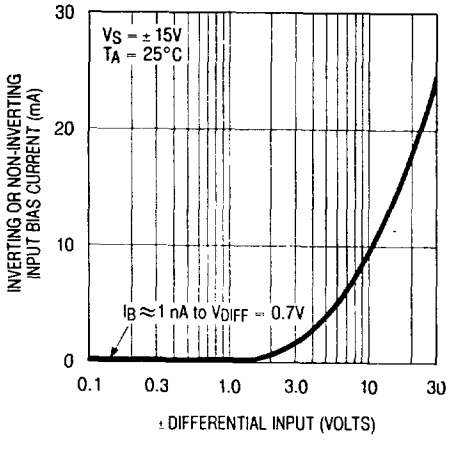
Input Bias and Offset Current vs Temperature



Input Bias Current Over the Common Mode Range

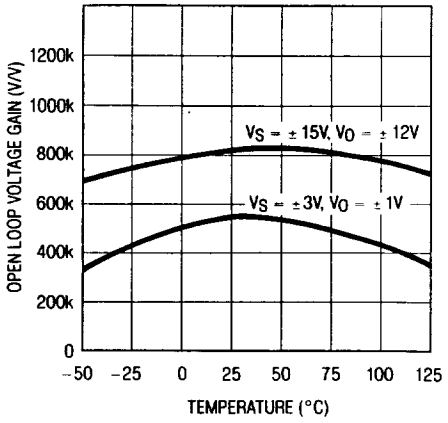


Input Bias Current vs. Differential Input Voltage

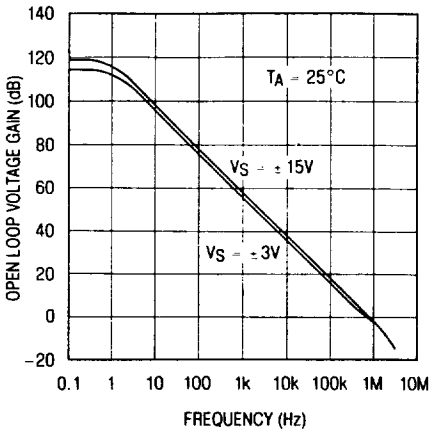


TYPICAL PERFORMANCE CHARACTERISTICS

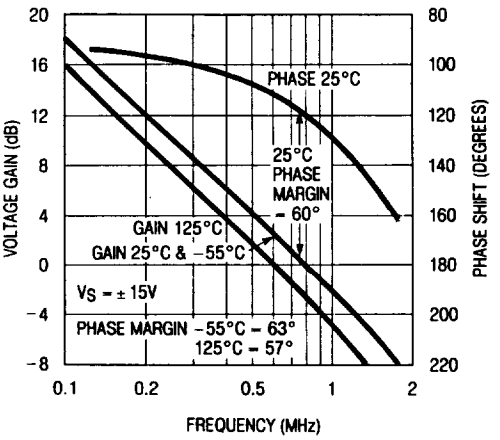
Open Loop Voltage Gain vs Temperature



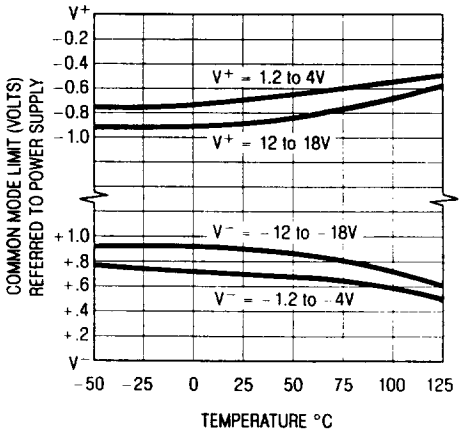
Open Loop Voltage Gain Frequency Response



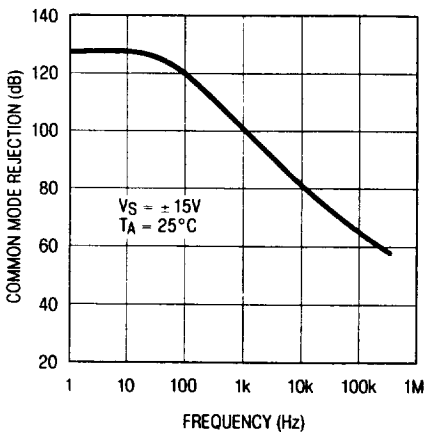
Gain, Phase Shift vs. Frequency



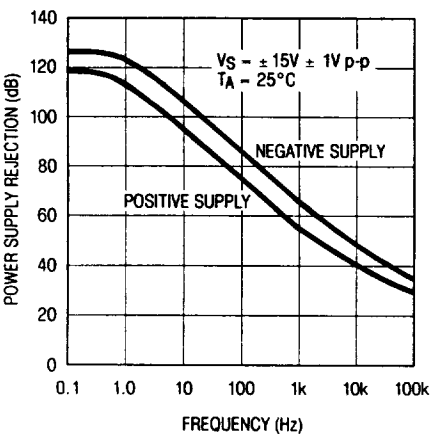
Common Mode Limit vs Temperature



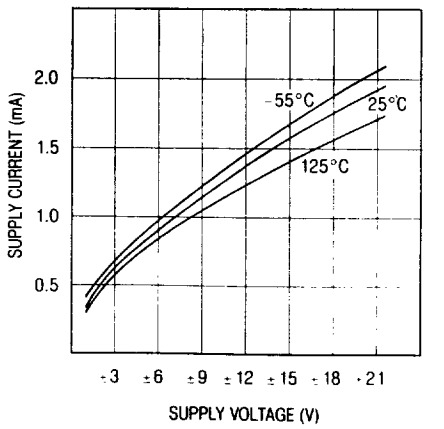
Common Mode Rejection Ratio vs Frequency



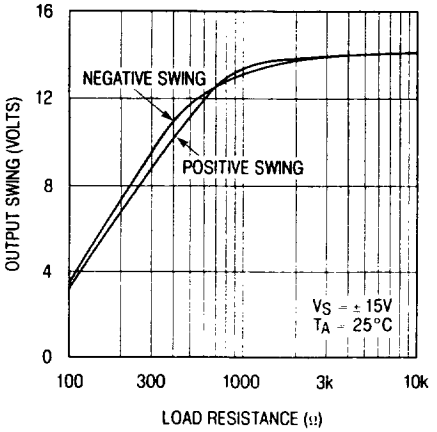
Power Supply Rejection Ratio vs Frequency



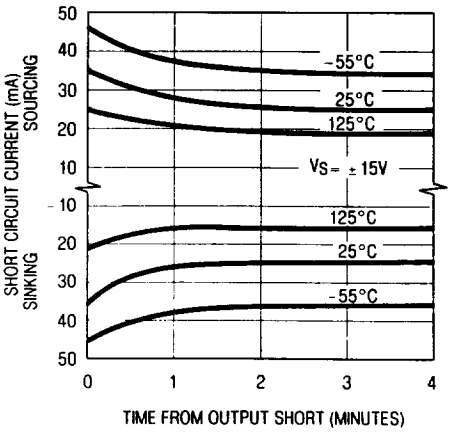
Supply Current vs Supply Voltage



Output Swing vs. Load Resistance

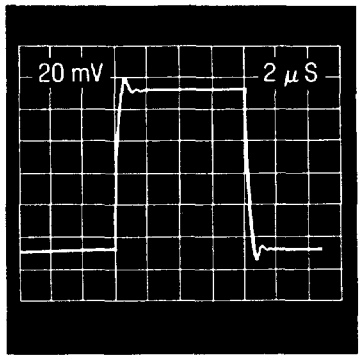


Output Short Circuit Current vs Time



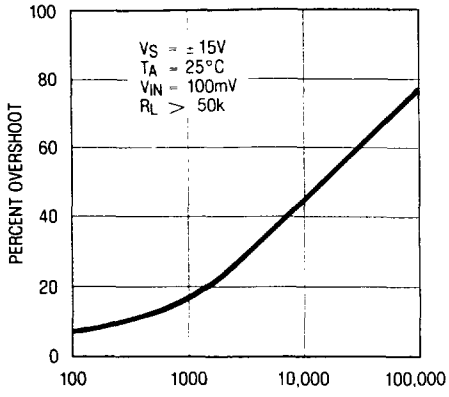
TYPICAL PERFORMANCE CHARACTERISTICS

Small Signal Transient Response

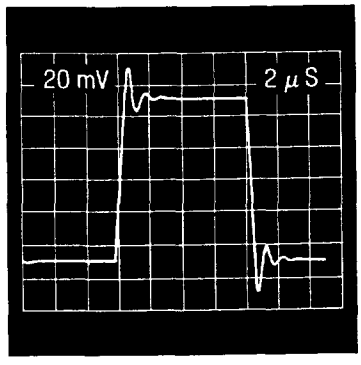


A_V = +1, C_L = 50pF

Voltage Follower Overshoot vs Capacitive Load

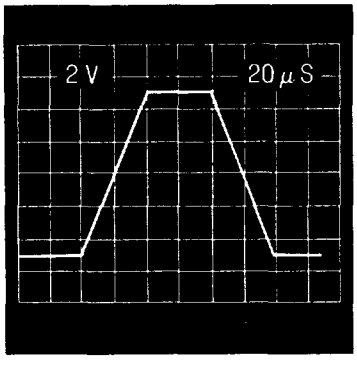


Small Signal Transient Response

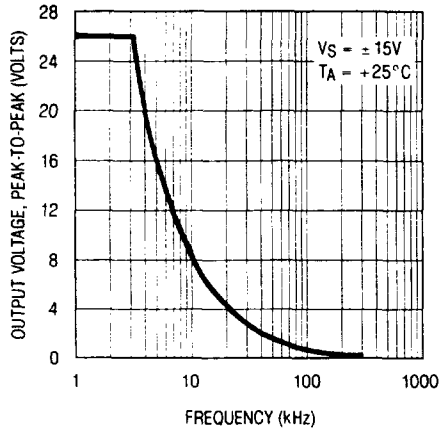


A_V = +1, C_L = 1000pF

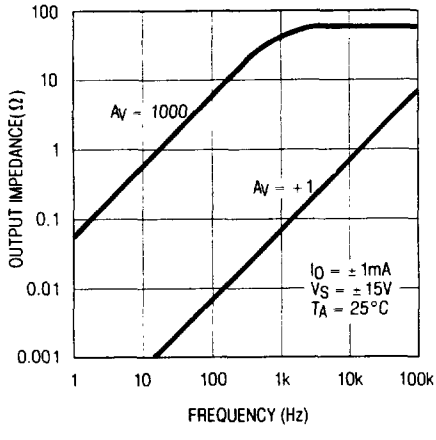
Large Signal Transient Response



Maximum Undistorted Output vs. Frequency



Closed Loop Output Impedance



APPLICATIONS INFORMATION

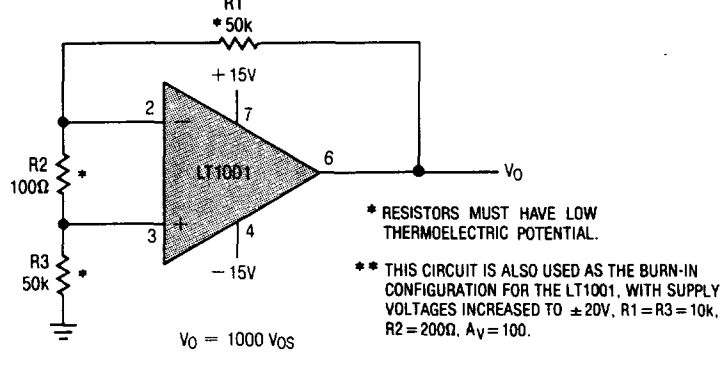
Application Notes and Test Circuits

The LT1001 series units may be inserted directly into OP-07, OP-05, 725, 108A or 101A sockets with or without removal of external frequency compensation or nulling components. The LT1001 can also be used in 741, LF156 or OP-15 applications provided that the nulling circuitry is removed.

The LT1001 is specified over a wide range of power supply voltages from ±3V to ±18V. Operation with lower supplies is possible down to ±1.2V (two Ni-Cad batteries). However, with ±1.2V supplies, the device is stable only in closed loop gains of +2 or higher (or inverting gain of one or higher).

Unless proper care is exercised, thermocouple effects caused by temperature gradients across dissimilar metals at the contacts to the input terminals, can exceed the inherent drift of the amplifier. Air currents over device leads should be minimized, package leads should be short, and the two input leads should be as close together as possible and maintained at the same temperature.

Test Circuit for Offset Voltage and its Drift with Temperature



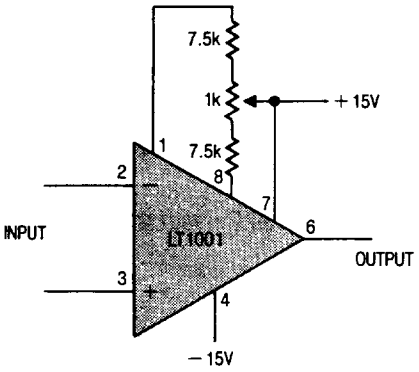
* RESISTORS MUST HAVE LOW THERMOELECTRIC POTENTIAL.

** THIS CIRCUIT IS ALSO USED AS THE BURN-IN CONFIGURATION FOR THE LT1001, WITH SUPPLY VOLTAGES INCREASED TO ±20V, R1 = R3 = 10k, R2 = 200Ω, A_V = 100.

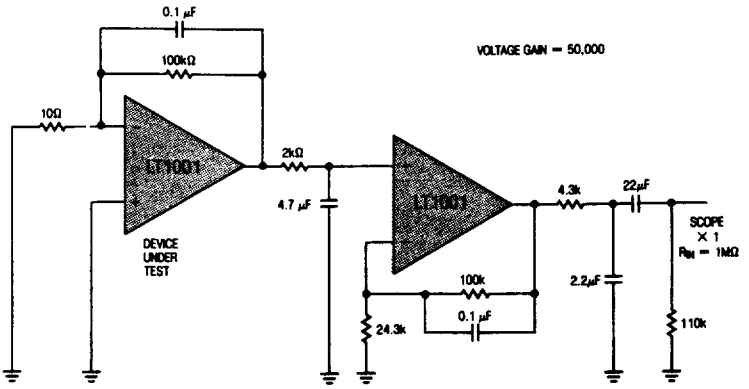
Offset Voltage Adjustment

The input offset voltage of the LT1001, and its drift with temperature, are permanently trimmed at wafer test to a low level. However, if further adjustment of Vos is necessary, nulling with a 10k or 20k potentiometer will not degrade drift with temperature. Trimming to a value other than zero creates a drift of (Vos/300) μV/°C, e.g. if Vos is adjusted to 300 μV, the change in drift will be 1 μV/°C. The adjustment range with a 10k or 20k pot is approximately $\pm$2.5mV. If less adjustment range is needed, the sensitivity and resolution of the nulling can be improved by using a smaller pot in conjunction with fixed resistors. The example below has an approximate null range of $\pm$ 100 μV.

Improved Sensitivity Adjustment



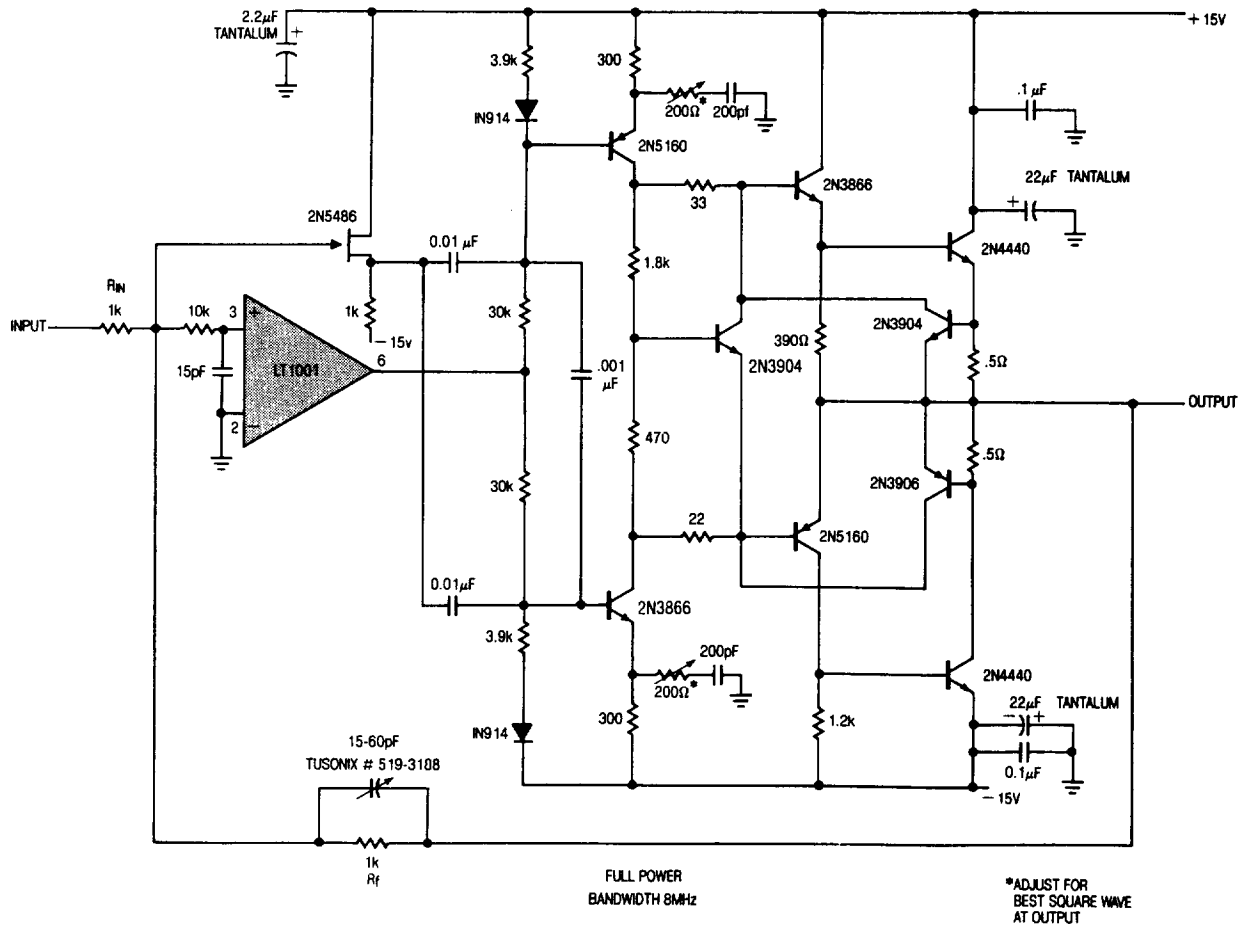
0.1Hz to 10Hz Noise Test Circuit



(Peak to Peak noise measured in 10 Sec interval)

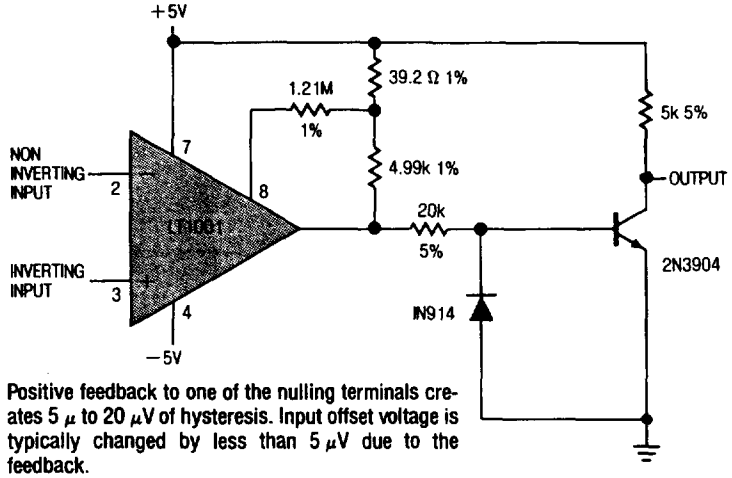
The device under test should be warmed up for three minutes and shielded from air currents.

DC Stabilized 1000v/μsec Op Amp



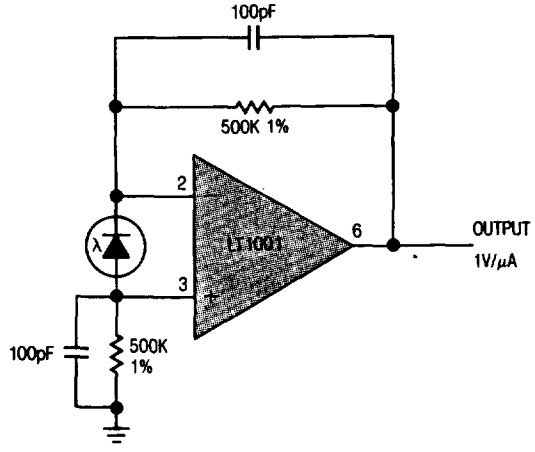
TYPICAL APPLICATIONS

Microvolt Comparator with TTL Output

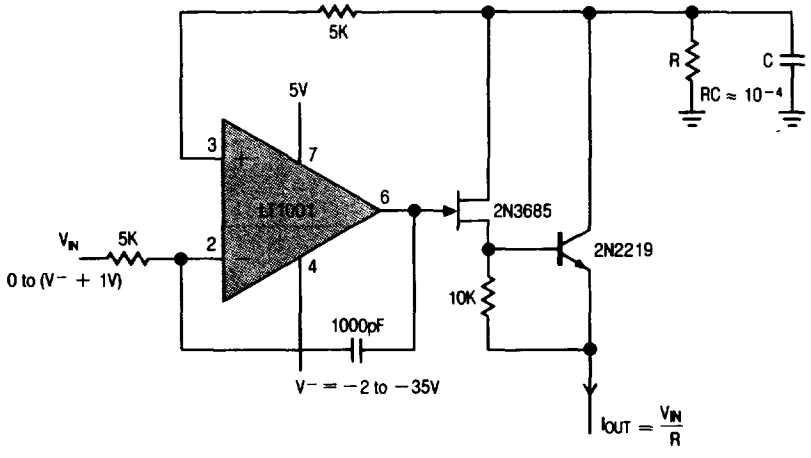


Positive feedback to one of the nulling terminals creates 5 μ to 20 μV of hysteresis. Input offset voltage is typically changed by less than 5 μV due to the feedback.

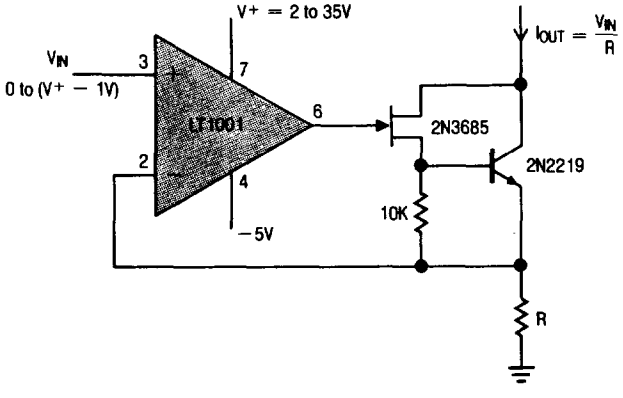
Photodiode Amplifier



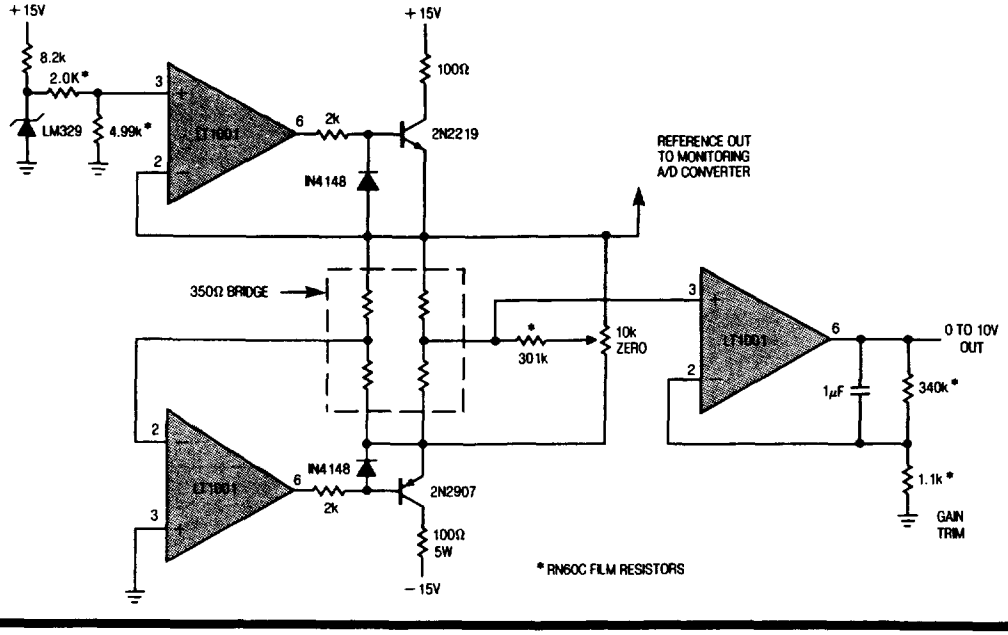
Precision Current Source



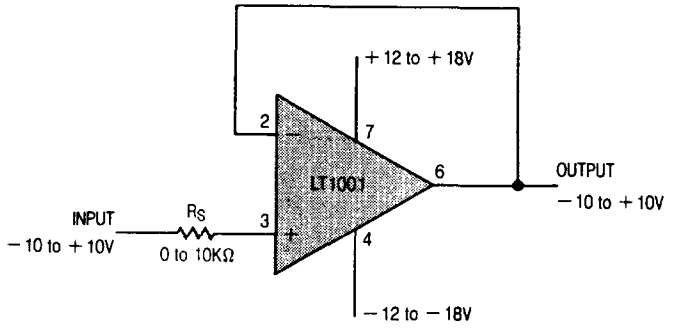
Precision Current Sink



Strain Gauge Signal Conditioner with Bridge Excitation



Large Signal Voltage Follower With 0.001% Worst-Case Accuracy



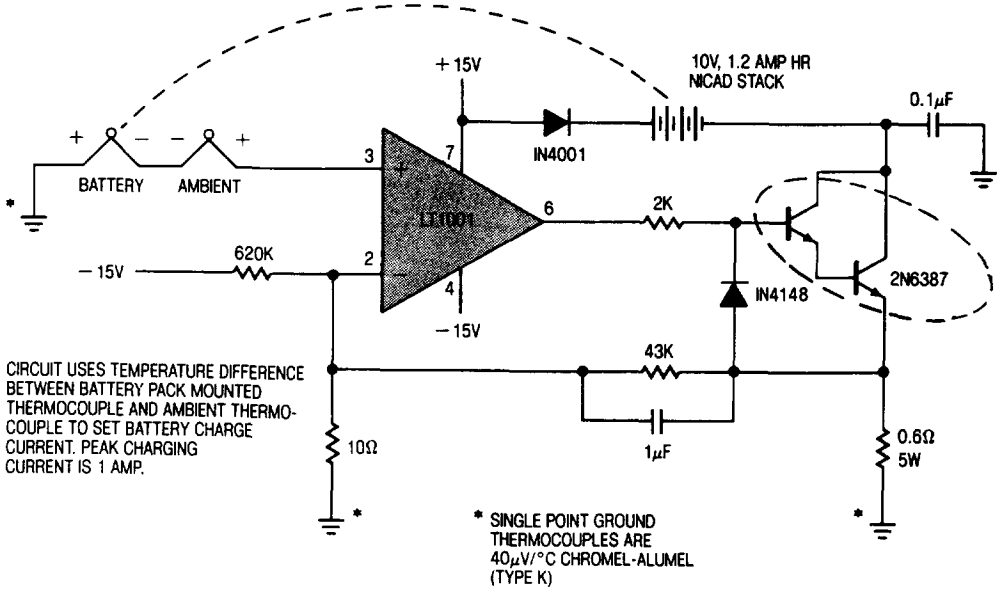
The voltage follower is an ideal example illustrating the overall excellence of the LT1001. The contributing error terms are due to offset voltage, input bias current, voltage gain, common-mode and power-supply rejections. Worst-case summation of guaranteed specifications is tabulated below.

OUTPUT ACCURACY

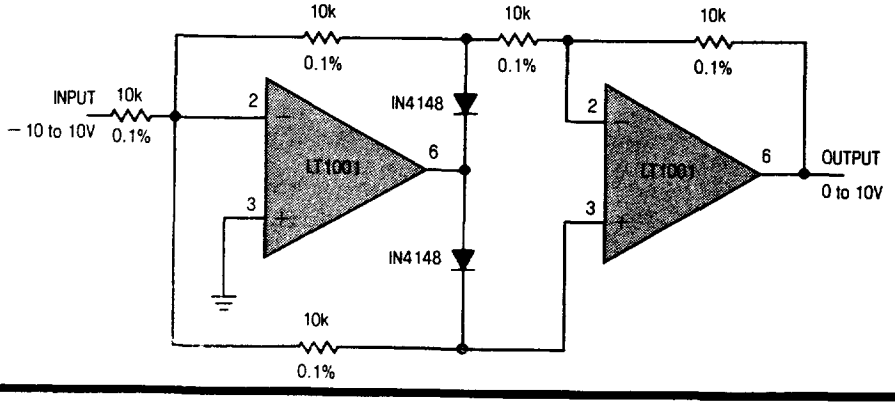
Error	LT1001AM /883 25°C Max.	LT1001C 25°C Max.	LT1001AM /883 −55 to 125°C Max.	LT1001C 0 to 70°C Max.
Offset Voltage	15μV	60μV	60μV	110μV
Bias Current	20μV	40μV	40μV	55μV
Common-Mode Rejection	20μV	30μV	30μV	50μV
Power Supply Rejection	18μV	30μV	36μV	42μV
Voltage Gain	22μV	25μV	33μV	40μV
Worst-case Sum	95μV	185μV	199μV	297μV
Percent of Full Scale (=20V)	0.0005%	0.0009%	0.0010%	0.0015%

2

Thermally Controlled Nicad Charger

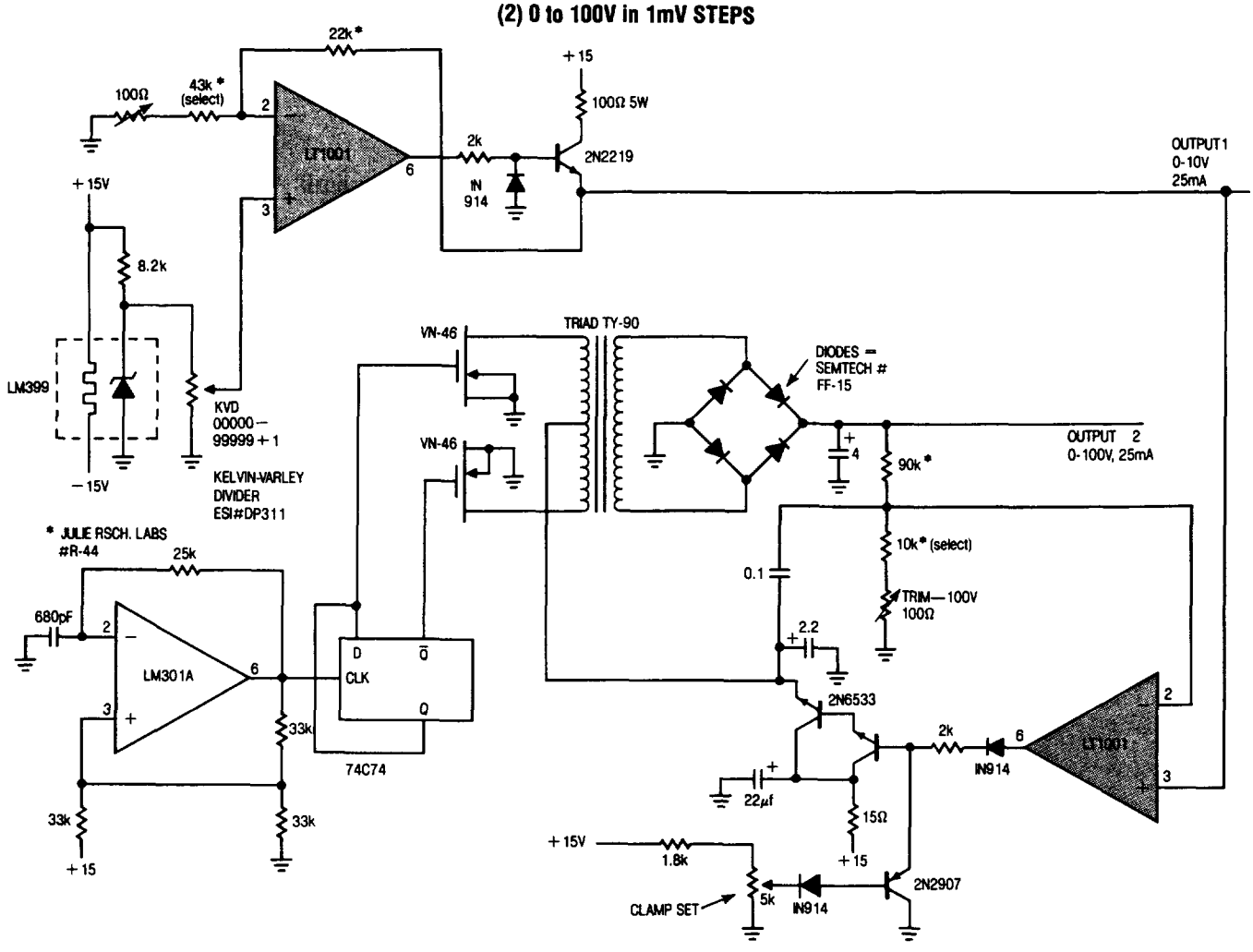


Precision Absolute Value Circuit



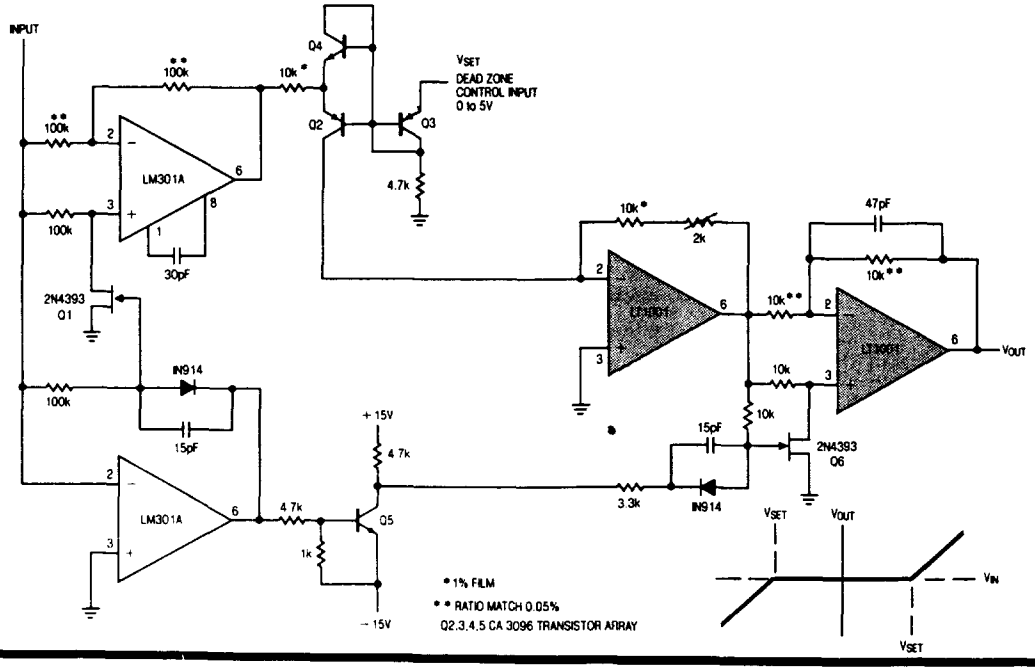
Precision Power Supply with Two Outputs

(1) 0 to 10V in 100μV STEPS
(2) 0 to 100V in 1mV STEPS

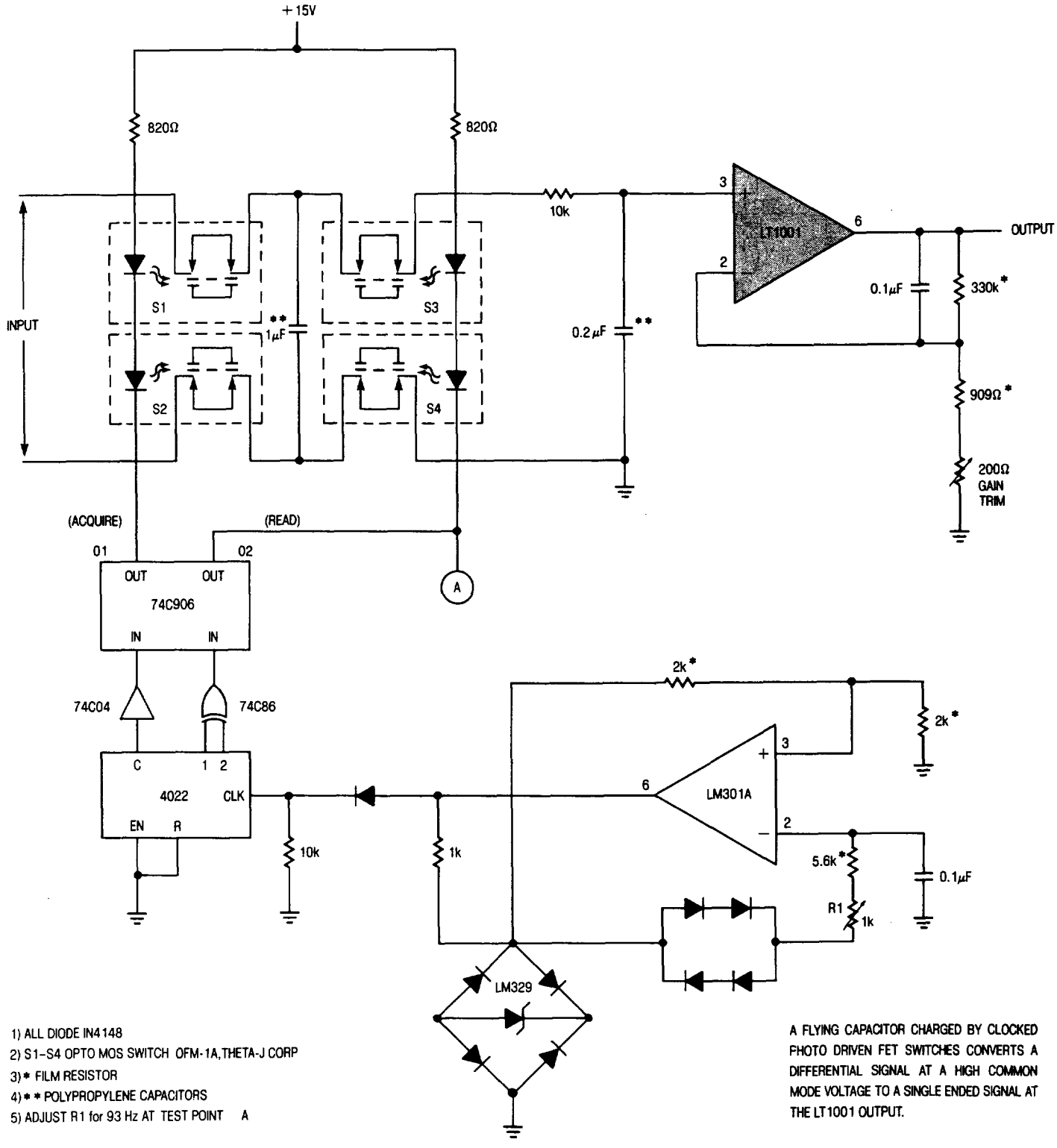


Dead Zone Generator

BIPOLAR SYMMETRY IS EXCELLENT BECAUSE ONE DEVICE, Q2, SETS BOTH LIMITS



Instrumentation Amplifier with ±300V Common Mode Range and CMRR > 150dB

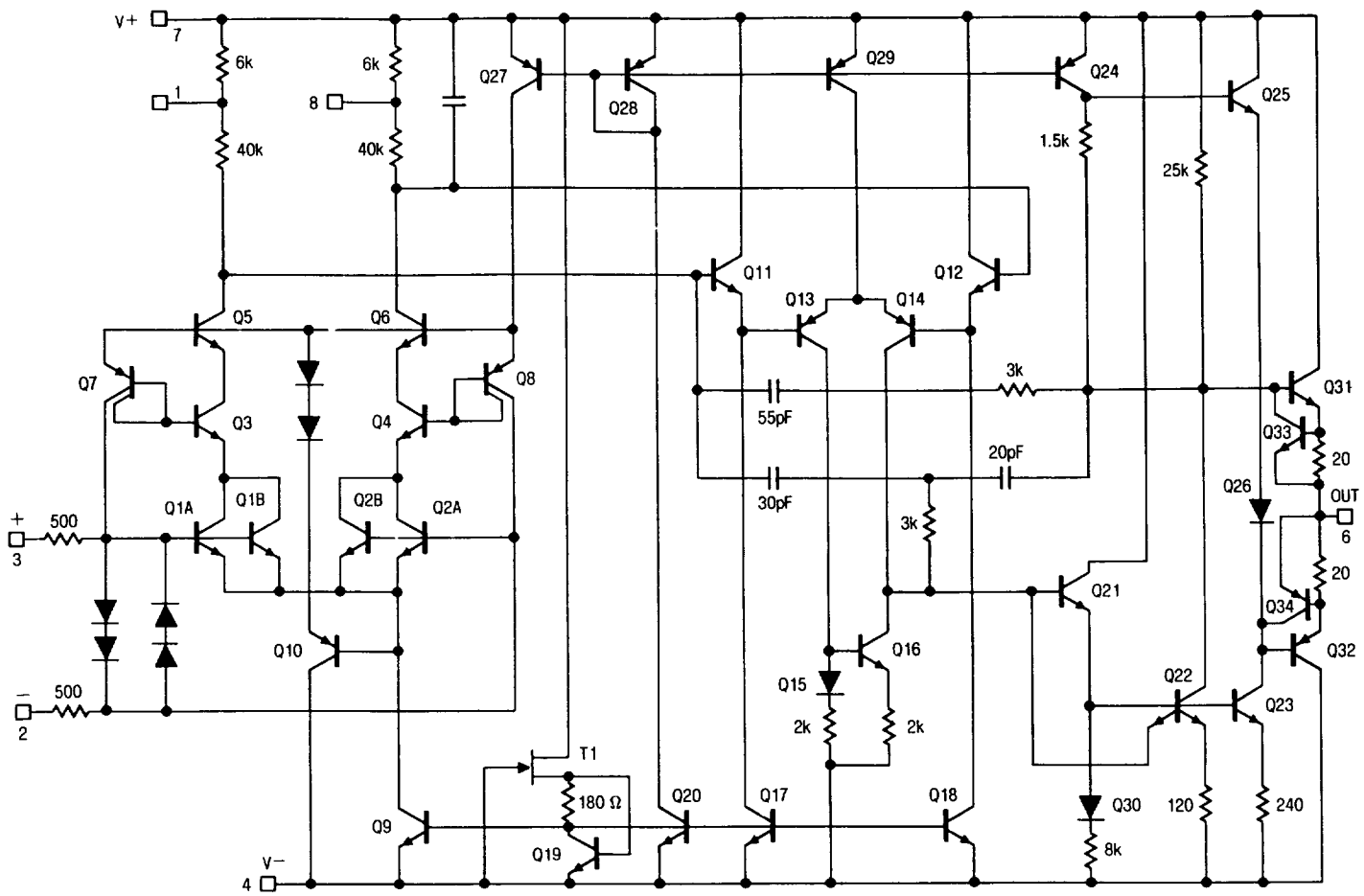


1) ALL DIODE IN4148
2) S1–S4 OPTO MOS SWITCH OFM-1A, THETA-J CORP
3) * FILM RESISTOR
4) * * POLYPROPYLENE CAPACITORS
5) ADJUST R1 for 93 Hz AT TEST POINT A

A FLYING CAPACITOR CHARGED BY CLOCKED PHOTO DRIVEN FET SWITCHES CONVERTS A DIFFERENTIAL SIGNAL AT A HIGH COMMON MODE VOLTAGE TO A SINGLE ENDED SIGNAL AT THE LT1001 OUTPUT.

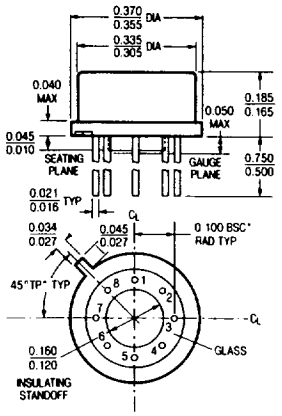
SCHEMATIC DIAGRAM

LT1001 Schematic Diagram



PACKAGE DESCRIPTION

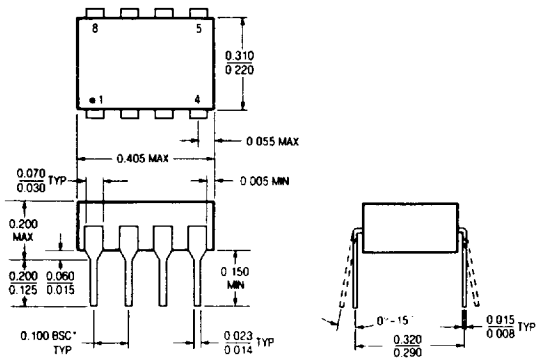
H Package
Metal Can



NOTE: DIMENSIONS IN INCHES

$T_j$max	Θ_{ja}	Θ_{jc}
150°C	150°C/W	45°C/W

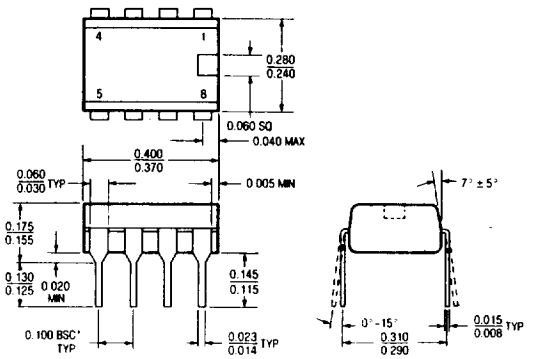
J8 Package
8 Lead Hermetic Dip



NOTE: DIMENSIONS IN INCHES UNLESS OTHERWISE NOTED
*LEADS WITHIN 0.007 OF TRUE POSITION (TP) AT GAUGE PLANE

$T_j$max	Θ_{ja}
150°C	100°C/W

N8 Package
8 Lead Plastic



NOTE: DIMENSIONS IN INCHES UNLESS OTHERWISE NOTED.
LEADS WITHIN 0.007 OF TRUE POSITION (TP) AT GAUGE PLANE.

$T_j$max	Θ_{ja}
100°C	130°C/W